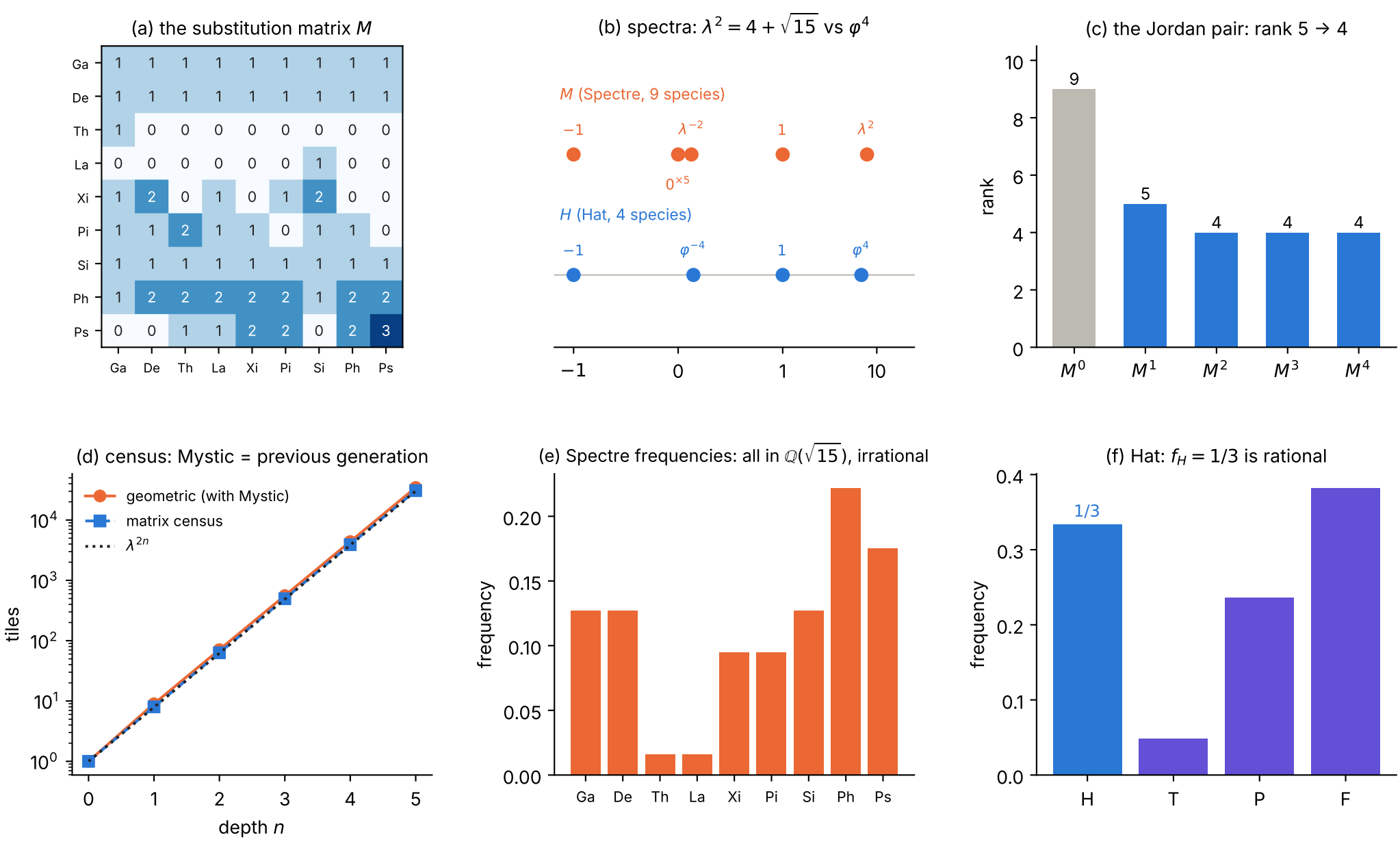
(a) the substitution matrix $M$
Ga 1 1 1 1 1 1 1 1 1
De 1 1 1 1 1 1 1 1 1
Th 1 0 0 0 0 0 0 0 0
La 0 0 0 0 0 0 1 0 0
Xi 1 2 0 1 0 1 2 0 0
Pi 1 1 2 1 1 0 1 1 0
Si 1 1 1 1 1 1 1 1 1
Ph 1 2 2 2 2 2 1 2 2
Ps 0 0 1 1 2 2 0 2 3
Ga De Th La Xi Pi Si Ph Ps
(b) spectra: $\lambda^2 = 4 + \sqrt{15}$ vs $\varphi^4$
$M$ (Spectre, 9 species)
$-1$ $\lambda^{-2}$ $1$ $\lambda^2$
$0^{\times 5}$
$H$ (Hat, 4 species)
$-1$ $\varphi^{-4}$ $1$ $\varphi^4$
$-1$ 0 1 10
(c) the Jordan pair: rank 5 → 4
rank
0 2 4 6 8 10
9 5 4 4 4
$M^0$ $M^1$ $M^2$ $M^3$ $M^4$
(d) census: Mystic = previous generation
geometric (with Mystic)
matrix census
$\lambda^{2n}$
tiles
$10^0$ $10^1$ $10^2$ $10^3$ $10^4$
0 1 2 3 4 5
depth $n$
(e) Spectre frequencies: all in $\mathbb{Q}(\sqrt{15})$, irrational
frequency
0.00 0.05 0.10 0.15 0.20
Ga De Th La Xi Pi Si Ph Ps
(f) Hat: $f_H = 1/3$ is rational
frequency
0.0 0.1 0.2 0.3 0.4
1/3
H T P F